

Beyond PII Removal: Risk-Budgeted Generalization for LLM-Ready Sensitive Text

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Abstract

1 Sensitive text corpora are valuable for training, retrieval, evaluation, and
 2 post-training adaptation of foundation-model systems, but common trans-
 3 formation pipelines still focus on direct personally identifying information
 4 (PII) such as names, emails, phone numbers, and addresses. This span-
 5 level view is necessary but incomplete: a record with no explicit name
 6 may still reveal sensitive attributes through quasi-identifiers such as exact
 7 dates, fine-grained locations, rare occupations, schools, medical or legal
 8 facts, family relations, citizenship cues, or combinations of ordinary details.
 9 We introduce Risk-Budgeted Quasi-Identifier Generalization (RB-QIG), a
 10 lightweight and auditable transformation layer that scores quasi-identifiers
 11 by privacy risk and utility importance, applies the least destructive step
 12 in a pre-specified generalization ladder, and stops when a document-level
 13 risk budget is met. On a controlled synthetic benchmark, RB-QIG balanced
 14 reduces risk-weighted leakage to 23.6% [22.9, 24.5] while preserving 71.7%
 15 [70.8, 72.4] of utility facts; blanket quasi-identifier redaction removes leak-
 16 age but preserves only 43.3% [39.9, 46.8] of utility facts. On a 100-record
 17 English RAT-Bench pilot, direct redaction leaves 78.0% [73.4, 82.5] LLM-
 18 attacker risk-weighted leakage and a naive LLM sanitizer leaves 46.0%
 19 [40.1, 52.0], while RB-QIG balanced reduces leakage to 5.7% [3.2, 8.8]. A
 20 blind public stress test similarly shows large risk reduction relative to direct
 21 redaction, although public semantic utility is tied with blanket redaction.
 22 RB-QIG is not an anonymization guarantee. It is evidence that responsi-
 23 ble data transformation for foundation-model pipelines should measure
 24 what capable models can still infer, not only which explicit PII spans were
 25 removed.

1 Introduction

27 Sensitive text is central to many data sources that would make foundation-model systems
 28 more useful: clinical notes, legal decisions, financial records, support logs, internal work-
 29 place messages, student advising records, and user-assistant conversations. Such data can
 30 support retrieval-augmented generation, model evaluation, safety auditing, preference-data
 31 construction, and domain adaptation. The same data also contains details that people did
 32 not intend to make public. Responsible data enablement therefore often begins with de-
 33 identification: remove names, replace emails, mask phone numbers, and redact addresses.

34 This direct-identifier pipeline is important, but it is not sufficient for LLM-ready text. A
 35 sentence such as “a 34-year-old oncology infusion pharmacist at a small startup in Asheville
 36 called about a brother’s rare condition after an April 2025 surgery” may contain no name
 37 or email after redaction, yet still expose a highly specific profile. Traditional privacy work
 38 formalized this problem through linkage and quasi-identifiers [Sweeney \(2002\)](#); [Machanava-](#)
 39 [jjhala et al. \(2007\)](#); [Li et al. \(2007\)](#); [Narayanan & Shmatikov \(2008\)](#). Modern language models
 40 make the problem more acute because an attacker need not match exact database columns:
 41 LLMs can combine paraphrase recognition, world knowledge, and contextual inference
 42 to recover sensitive attributes [Staab et al. \(2024\)](#). Recent text-anonymization benchmarks
 43 similarly argue that anonymization should be evaluated by residual re-identification risk,
 44 not only by span recall [Krčo et al. \(2026\)](#).

The practical question is therefore not just “did the tool remove obvious PII?” but “what can a capable model still infer, and how much useful meaning was preserved?” A conservative answer is to redact every quasi-identifier: occupations become [OCCUPATION], locations become [LOCATION], dates become [DATE], and diagnoses become [MEDICAL]. This can reduce leakage, but it also destroys much of the information that made the corpus useful. The opposite extreme—direct PII removal only—preserves utility but leaves large residual inference risk.

We study the space between these extremes through Risk-Budgeted Quasi-Identifier Generalization (RB-QIG). RB-QIG does not train a model and does not rewrite whole documents. It takes candidate quasi-identifier spans, assigns each a privacy-risk score and a utility-importance score, and applies span-level replacements from a small ladder: keep, generalize once, generalize further, or redact. The algorithm greedily chooses the next edit that gives the largest risk reduction per unit of utility loss, stopping when a document-level risk budget is satisfied. The resulting audit log records which fact was generalized, which replacement was used, and why.

Our contributions are: (i) a quasi-identifier taxonomy for LLM-ready sensitive text; (ii) a risk-budgeted generalization policy that operationalizes an inspectable privacy-utility tradeoff without training or fine-tuning; (iii) an evaluation protocol combining deterministic leakage metrics, paired bootstrap intervals, cached LLM attackers, and cached LLM utility judges; and (iv) an analysis showing that residual leakage often comes from relational or lexical implications rather than missed exact spans. Our strongest supported claim is narrow but important: direct PII redaction and naive LLM sanitization leave substantial residual quasi-identifier risk, while RB-QIG sharply reduces that measured risk and exposes a useful controlled privacy-utility frontier.

2 Background and Threat Model

From tabular quasi-identifiers to text. Classical privacy work showed that attributes such as ZIP code, gender, and birth date can link a record to a person even when names are absent [Sweeney \(2002\)](#). l -diversity and t -closeness refined this view by showing that indistinguishability alone may not protect sensitive attributes [Machanavajjhala et al. \(2007\)](#); [Li et al. \(2007\)](#). Differential privacy offers a different guarantee for statistical releases [Dwork et al. \(2006\)](#), but it does not directly solve the instance-level text-transformation problem studied here: preserve a useful text record while reducing inferable identifiers inside that record.

Text de-identification. Applied de-identification systems often target named entities and regulated identifier categories. Clinical de-identification has a long history of shared tasks and neural sequence-labeling systems [Uzuner et al. \(2007\)](#); [Dernoncourt et al. \(2017\)](#). Practical tools such as Presidio support PII detection, masking, and replacement [Microsoft \(2026\)](#); specialized privacy filters provide another local PII-masking building block [OpenAI \(2026\)](#). These systems are useful, but they are typically optimized for identifiable spans rather than the residual inference risk created by combinations of demographic, relational, narrative, and domain facts. TAB provides a legal-domain anonymization benchmark [Pilán et al. \(2022\)](#), and RAT-Bench directly emphasizes LLM-based re-identification risk from direct and indirect identifiers [Krčo et al. \(2026\)](#).

LLM privacy leakage. LLMs introduce two complementary privacy concerns. Training data may be memorized or extractable under some conditions [Carlini et al. \(2021; 2023\)](#). Separately, even when a private fact is not explicit, a model may infer it from surrounding text. Staab et al. show that LLMs can infer personal attributes from ordinary text in their setting [Staab et al. \(2024\)](#). RB-QIG focuses on the second concern at the transformation stage: before using sensitive text for retrieval, evaluation, or post-training, reduce what a capable model can infer from the transformed record.

Threat model. The attacker receives transformed text and, in the LLM-attacker setting, a list of benchmark attributes to infer. The attacker returns structured guesses with short

Table 1: Quasi-identifier categories used by RB-QIG. The examples are illustrative; the implementation stores type, risk, utility, and a replacement ladder for each candidate span.

Category	Example precise value	Safer generalized value
Date/age	April 17, 2025; 34	month/year; 30s
Location	Asheville, NC	city in the southeastern U.S.
Occupation	oncology infusion pharmacist	clinical pharmacy professional
Institution	named school or employer	university; healthcare company
Medical/legal fact	rare condition or case event	broader condition/event type
Family relation	younger brother under guardianship	dependent family member
Demographic cue	citizenship or heritage phrase	broader demographic placeholder
Narrative cue	unique life event	coarse role-preserving description

evidence. We report exact recovery, coarse recovery, record compromise, and risk-weighted leakage. We use only synthetic profiles and public benchmarks; we do not attempt to identify real people. This threat model does not cover arbitrary auxiliary databases or determined human investigators. It is a reproducible stress test for whether a transformation leaves sensitive attributes inferable.

We distinguish three evaluation settings. In the *oracle/controlled* setting, ground-truth quasi-identifier spans are known from synthetic data. In the *target-aware public* setting, RAT-Bench target attributes are used to find value-revealing spans; this isolates transformation quality when the sensitive attributes of interest are known. In the *blind public* setting, the extractor does not see target attributes at extraction time and is scored against benchmark annotations afterward. The blind setting is closest to deployment and is where our evidence is most caveated.

3 Risk-Budgeted Quasi-Identifier Generalization

3.1 Taxonomy and Generalization Ladders

RB-QIG uses typed candidate spans. The taxonomy is intentionally broad because LLM inference often uses facts that are not direct PII. Table 1 summarizes the categories used in the prototype.

A generalization ladder is a short ordered list of replacements. For example, “Boise, Idaho” may map to “a city in the U.S. Mountain West,” then “a city in the western United States,” and finally [LOCATION]. The ladder is not learned; it is a transparent policy artifact. This matters for sensitive data owners because they may want to inspect, revise, or prohibit particular replacement types before releasing transformed text.

3.2 Risk and Utility Scores

Each quasi-identifier q_i receives a privacy-risk score $r_i \in [0, 5]$ and a utility-importance score $u_i \in [0, 5]$. Risk reflects specificity, rarity, sensitivity, and linkability. Utility reflects whether the fact is needed for the benign task: diagnosis category may be important in a clinical triage note, while an exact date of birth is usually not. The current prototype uses transparent heuristics and benchmark metadata rather than learned risk models. This is a limitation, but it also makes the method auditable.

The document risk score is

$$R(x) = \sum_i r_i + 0.5 \binom{|\{i : r_i \geq 3\}|}{2},$$

where the second term approximates combination risk from multiple linkable high-risk attributes. Candidate edits are evaluated by their risk drop divided by utility loss. RB-QIG greedily applies the edit with the best ratio until $R(x)$ falls below the budget or no edits remain. All edits are span replacements, so the system avoids unconstrained paraphrase hallucination.

3.3 Baselines

We compare against five baselines. *None* leaves the record unchanged. *Direct redaction* removes direct identifiers using regexes and benchmark-provided direct spans. *Naive LLM direct* asks an LLM sanitizer to remove personally identifying information while preserving meaning. *Blanket QI redaction* replaces every quasi-identifier with a typed placeholder. *Presidio-style pattern redaction* is a practical pattern baseline included as a lower-bound diagnostic rather than a full production Presidio deployment. RB-QIG is evaluated with strict, balanced, and utility-oriented budgets in the synthetic setting, with the balanced policy as the main public method.

4 Experiments

Synthetic control. We generate 100 synthetic records across medical, legal, financial, workplace, and education-support domains. Each record has direct identifiers, quasi-identifiers, ground-truth attributes, a task label, and utility facts. Because oracle quasi-identifier spans are known, this setting isolates the transformation tradeoff: given the sensitive spans, does risk-budgeted generalization preserve more useful information than blanket redaction?

RAT-Bench pilot. We evaluate 100 English difficulty-1 RAT-Bench records Krčo et al. (2026). The target-aware version uses benchmark target attributes to identify value-revealing spans and variants. This is a controlled transformation benchmark, not a deployment claim. We also run a blind extractor over the same 100 records and score it against benchmark quasi-identifiers. A deterministic backstop for common demographic, citizenship, education, employment, marital, and race/ethnicity cues improves coverage without extra API calls.

TAB legal screen. We run a 30-document TAB ECHR legal-domain screen Pilán et al. (2022). The TAB experiment is intentionally bounded and deterministic. It tests whether the residual-risk story transfers to another domain, but it is not used as a headline utility win because the small legal-task utility screen is inconclusive.

LLM attacker and utility judge. For RAT-Bench, a cached LLM attacker receives transformed text and requested attribute names, then returns structured guesses with evidence. A cached LLM utility judge scores label preservation, fact preservation, and semantic utility without rewarding preserved direct identifiers. All API calls use synthetic or public benchmark records, are cached, and cost approximately \$1.85 fresh-equivalent in the current workspace artifacts.

Metrics. Privacy metrics are direct-identifier leakage, exact attribute leakage, coarse attribute leakage, record compromise, and risk-weighted leakage. Utility metrics are utility-fact preservation, label preservation, semantic utility, annotation-derived quasi-identifier specificity, and token change. We report paired bootstrap 95% confidence intervals for headline comparisons.

5 Results and Analysis

Controlled privacy-utility frontier. Table 2 shows the cleanest evidence for the method. Direct redaction preserves utility but leaves all quasi-identifiers. Blanket quasi-identifier redaction eliminates measured leakage but preserves only 43.3% of utility facts. RB-QIG balanced preserves 71.7% of utility facts while reducing risk-weighted leakage from 100.0% to 23.6%. The paired bootstrap advantage in utility-fact preservation over blanket redaction is +28.3 points [25.2, 31.6]. This supports the controlled claim that risk-budgeted generalization can occupy a useful middle point between direct redaction and blanket redaction.

Semantic utility is more modest. The LLM utility judge in Table 4 is more forgiving than the synthetic fact metric because broad placeholders can still preserve some document

Table 2: Synthetic controlled benchmark. Brackets show bootstrap 95% confidence intervals.

Method	Risk-weighted leak	Utility facts	Token change
None	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
Direct	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	13.5 [12.6, 14.3]
Blanket QI	0.0 [0.0, 0.0]	43.3 [39.9, 46.8]	63.7 [62.9, 64.4]
RB-QIG strict	16.1 [15.7, 16.4]	48.3 [45.3, 51.4]	59.3 [58.7, 60.0]
RB-QIG balanced	23.6 [22.9, 24.5]	71.7 [70.8, 72.4]	59.0 [58.3, 59.7]
RB-QIG utility	38.0 [37.5, 38.4]	95.0 [93.0, 96.8]	56.6 [55.8, 57.5]

Table 3: RAT-Bench LLM attacker results. Brackets show bootstrap 95% confidence intervals.

Method	Record compromise	Exact attr leak	Coarse attr leak	Risk-weighted leak
Direct	76.0 [67.0, 84.0]	74.9 [69.5, 80.0]	81.2 [76.8, 85.4]	78.0 [73.4, 82.5]
LLM direct	34.0 [25.0, 43.0]	48.7 [42.6, 54.6]	52.8 [46.5, 58.7]	46.0 [40.1, 52.0]
Blanket QI	3.0 [0.0, 7.0]	3.9 [1.5, 7.0]	7.3 [4.3, 10.8]	5.3 [2.7, 8.3]
RB-QIG balanced	2.0 [0.0, 5.0]	4.0 [1.6, 6.9]	8.8 [5.3, 12.8]	5.7 [3.2, 8.8]

179 meaning. Under this metric, RB-QIG balanced has only a modest semantic utility edge over
 180 blanket redaction: +2.6 points [0.6, 4.8]. On blind public RAT-Bench, the same judge finds
 181 no meaningful edge over blanket redaction. We therefore scope the large utility claim to
 182 controlled synthetic facts and report public utility as an unresolved limitation.

183 **Direct redaction leaves high residual LLM risk.** Table 3 shows that direct redaction is
 184 insufficient on the RAT-Bench pilot. After direct redaction, the LLM attacker still recovers
 185 many target attributes: 78.0% risk-weighted leakage. The naive LLM sanitizer improves
 186 over direct redaction but still leaves 46.0% leakage. RB-QIG balanced reduces leakage to
 187 5.7%, a paired drop of 72.2 points [67.0, 77.4] relative to direct redaction and 40.2 points
 188 [33.1, 47.1] relative to the LLM sanitizer. RB-QIG and blanket redaction are statistically
 189 tied on this privacy metric, so the privacy claim is reduction relative to direct and naive
 190 redaction, not superiority over blanket redaction.

191 **Blind extraction is the deployment bottleneck.** Table 5 summarizes the blind public stress
 192 test. The raw blind extractor covers 72.8% of benchmark QI spans; a generic deterministic
 193 backstop raises coverage to 99.6% without additional API calls. Under the LLM attacker,
 194 blind-backstop RB-QIG balanced reduces risk-weighted leakage from 78.1% [73.3, 82.9] for
 195 direct redaction to 6.4% [3.9, 9.1], and is statistically tied with blanket redaction. Public
 196 semantic utility is also tied with blanket redaction, so the blind setting should be read as
 197 evidence for residual-risk reduction and as evidence that robust utility-preserving quasi-
 198 identifier extraction remains open.

199 **Cross-domain and practical-baseline diagnostics.** The TAB screen provides deterministic
 200 cross-domain support: direct redaction leaves 99.8% [99.4, 100.0] risk-weighted leakage,
 201 while RB-QIG balanced lowers it to 12.3% [9.2, 15.7]. Its legal-task utility screen is incon-
 202 clusive and therefore not used as a positive utility result. The Presidio-style pattern-only
 203 baseline gives a practical lower-bound diagnostic: on RAT-Bench it leaves 37.0% [28.0, 46.0]
 204 direct-identifier leakage and 85.0% [80.6, 89.0] deterministic risk-weighted leakage; on TAB
 205 it misses legal names and case identifiers, leaving 100.0% direct-identifier leakage. This
 206 makes the oracle-assisted direct baseline conservative rather than weak.

207 **Failure taxonomy.** Residual RB-QIG leaks are mostly relational or lexical variants: widow-
 208 hood from bereavement cues, sex from gendered language, education from school-name
 209 mentions, citizenship from non-citizen phrasing, and employment or armed-forces status
 210 from occupation context. These failures are not just missed exact spans; they are cues that a
 211 model can combine. This supports the broader argument that data transformation should
 212 evaluate inferability, not only span categories.

Table 4: Synthetic LLM utility judge results. Brackets show bootstrap 95% confidence intervals.

Method	Label preserved	LLM fact preservation	Semantic utility
Direct	100.0	75.3 [73.3, 77.3]	83.2 [81.8, 84.8]
Blanket QI	99.0	69.1 [66.6, 71.4]	76.8 [74.8, 78.4]
RB-QIG balanced	100.0	71.3 [69.6, 73.0]	79.4 [78.2, 80.6]

Table 5: Blind public RAT-Bench stress test. Brackets show bootstrap 95% confidence intervals.

Method	LLM risk-weighted leak	Semantic utility	LLM fact preservation
Direct	78.1 [73.3, 82.9]	86.2 [83.2, 88.8]	84.3 [81.2, 87.0]
Blanket QI	6.8 [4.0, 9.9]	62.6 [59.2, 66.0]	53.8 [49.5, 58.1]
RB-QIG balanced	6.4 [3.9, 9.1]	62.2 [58.8, 65.2]	55.5 [52.0, 58.7]

6 Limitations and Ethics

RB-QIG is not an anonymization guarantee and should not be used as a legal or compliance claim. The main public benchmark extraction is target-aware, so it evaluates transformation behavior under known benchmark attributes rather than fully blind deployment. Blind public and synthetic diagnostics suggest that utility-aware conversion and deterministic backstops help, but robust blind quasi-identifier extraction remains unresolved. LLM attackers and judges are low-cost automated evaluators and may under- or over-estimate real adversaries or utility needs. Our risk and utility scores are transparent heuristics, not universal privacy estimates. All experiments use synthetic or public benchmark data. We do not attempt to identify real people or infer private attributes about individuals.

7 Conclusion

Direct PII removal is an incomplete transformation for LLM-ready sensitive text. RB-QIG demonstrates that a small, auditable risk-budgeted generalization layer can dramatically reduce measured residual inference risk while preserving useful semantic content better than blanket redaction in controlled settings. The main open problem is robust blind quasi-identifier extraction and rewriting that preserves task-relevant utility while handling relational, lexical, and world-knowledge inference cues. More broadly, responsible data enablement should evaluate what capable models can still infer, not only what explicit PII spans were removed.

A Reproducibility Notes

The artifact package is generated from local scripts under `src/` and `public/synthetic` data only. The main synthetic control is `results/synthetic_100/`; the public RAT-Bench pilot is `results/ratbench_d1_api_100/`; and the blind public stress test is `results/ratbench_d1_blind_backstop_v2_budgetfix_api_100/`. Bootstrap intervals are generated by `src/bootstrap_results.py`, and paper tables by `src/make_paper_tables.py`. The qualitative appendix is generated by `src/make_qualitative_appendix.py`; headline claim checks are generated by `src/make_claim_audit.py`. Cached API usage in the workspace is approximately \$1.85 fresh-equivalent, and all API calls use public or synthetic benchmark records.

References

Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *USENIX Security Symposium*, 2021.

- 246 Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and
247 Chiyuan Zhang. Quantifying memorization across neural language models. In *International Conference on Learning Representations*, 2023.
248
- 249 Franck Dernoncourt, Ji Young Lee, Özlem Uzuner, and Peter Szolovits. De-identification of
250 patient notes with recurrent neural networks. In *Journal of the American Medical Informatics Association*, volume 24, pp. 596–606, 2017.
251
- 252 Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to
253 sensitivity in private data analysis. In *Theory of Cryptography Conference*, pp. 265–284, 2006.
- 254 Nataša Krčo, Zexi Yao, Matthieu Meeus, and Yves-Alexandre de Montjoye. RAT-Bench: A
255 comprehensive benchmark for text anonymization, 2026. URL [https://arxiv.org/abs/](https://arxiv.org/abs/2602.12806)
256 [2602.12806](https://arxiv.org/abs/2602.12806).
- 257 Ninghui Li, Tiancheng Li, and Suresh Venkatasubramanian. t-closeness: Privacy beyond
258 k-anonymity and l-diversity. In *IEEE International Conference on Data Engineering*, pp.
259 106–115, 2007.
- 260 Ashwin Machanavajjhala, Daniel Kifer, Johannes Gehrke, and Muthuramakrishnan Venkata-
261 subramaniam. l-diversity: Privacy beyond k-anonymity. *ACM Transactions on Knowledge*
262 *Discovery from Data*, 1(1), 2007.
- 263 Microsoft. Microsoft presidio: Data protection and de-identification sdk, 2026. URL [https:](https://microsoft.github.io/presidio/)
264 [//microsoft.github.io/presidio/](https://microsoft.github.io/presidio/). Accessed 2026-06-27.
- 265 Arvind Narayanan and Vitaly Shmatikov. Robust de-anonymization of large sparse datasets.
266 In *IEEE Symposium on Security and Privacy*, pp. 111–125, 2008.
- 267 OpenAI. Introducing openai privacy filter, 2026. URL [https://openai.com/index/](https://openai.com/index/introducing-openai-privacy-filter/)
268 [introducing-openai-privacy-filter/](https://openai.com/index/introducing-openai-privacy-filter/). Accessed 2026-06-27.
- 269 Ildikó Pilán, Pierre Lison, Lilja Øvrelid, Anthi Papadopoulou, David Sánchez, and Montser-
270 rat Batet. The Text Anonymization Benchmark (TAB): A dedicated corpus and evaluation
271 framework for text anonymization, 2022. URL <https://arxiv.org/abs/2202.00443>.
- 272 Robin Staab, Mark Vero, Mislav Balunovic, and Martin Vechev. Beyond memorization:
273 Violating privacy via inference with large language models. In *International Conference on*
274 *Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.07298>.
- 275 Latanya Sweeney. k-anonymity: A model for protecting privacy. In *International Journal of*
276 *Uncertainty, Fuzziness and Knowledge-Based Systems*, volume 10, pp. 557–570, 2002.
- 277 Özlem Uzuner, Yuan Luo, and Peter Szolovits. Evaluating the state-of-the-art in automatic
278 de-identification. In *Journal of the American Medical Informatics Association*, volume 14, pp.
279 550–563, 2007.